



Phison Electronics Corporation

microSD 6.1 (PS8229) KIC TLC

Specification

(UHS-I)

Version 1.3

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Revision History

[illegible]

Overview

- **Flash Type**
 - BiCS3
 - BiCS4
 - BiCS5
 - BiCS6
- **Video Speed Class**
 - Up to V30
- **Bus Speed Mode**
 - UHS-I
- **Speed Class**
 - Class 10
 - A2
 - UHS-I, U1/U3
- **Power Consumption** ^{Note}
 - Power Up Current $\leq 250\mu\text{A}$
 - Standby Current $\leq 1000\mu\text{A}$
 - Read Current $\leq 400\text{mA}$
 - Write Current $\leq 400\text{mA}$
- **Advanced Flash Management**
 - ECC Correction
 - Static and Dynamic Wear Leveling
 - Bad Block Management
- **Write Protect with mechanical switch**
- **Supply Voltage 2.7 ~ 3.6V**
- **Temperature Range**
 - Operation: $-25^{\circ}\text{C} \sim 85^{\circ}\text{C}$
 - Storage: $-40^{\circ}\text{C} \sim 85^{\circ}\text{C}$
- **RoHS compliant**
- **EMI compliant**

NOTE: Please see Chapter 6.1 Power Consumption for details.

Performance Overview

Capacity	Class	UHS-I	VSC	APP PERF Class	Controller	Flash			TestMetrix IOPs test		TestMetrix Performance	
						Density	Process	Bit-per-cell	Read (IOPs)	Write (IOPs)	Read (MB/s)	Write (MB/s)
16GB	CL10	UHS-I (Grade 1)	V10	A2	PS8229	128Gbx1	BiCS3	TLC	9804	5831	100.91	41.13
32GB	CL10	UHS-I (Grade 1)	V10	A2	PS8229	256Gbx1	BiCS3	TLC	9804	5836	100.84	40.26
64GB	CL10	UHS-I (Grade 3)	V30	A2	PS8229	256Gbx2	BiCS3	TLC	9804	4282	100.84	78.74
128GB	CL10	UHS-I (Grade 3)	V30	A2	PS8229	256Gbx4	BiCS3	TLC	9541	3972	100.84	93.89
256GB	CL10	UHS-I (Grade 3)	V30	A2	PS8229	256Gbx8	BiCS3	TLC	9804	4563	100.84	94.12
32GB	CL10	UHS-I (Grade 1)	V10	A2	PS8229	256Gbx1	BiCS4	TLC	9804	6208	100.83	55.31
64GB	CL10	UHS-I (Grade 3)	V30	A2	PS8229	256Gbx2	BiCS4	TLC	9804	4717	100.83	94.94
128GB	CL10	UHS-I (Grade 3)	V30	A2	PS8229	256Gbx4	BiCS4	TLC	9804	6164	100.83	92.63
256GB	CL10	UHS-I (Grade 3)	V30	A2	PS8229	256Gbx8	BiCS4	TLC	9804	6570	100.91	95.36
64GB	CL10	UHS-I (Grade 1)	V10	A2	PS8229	512Gbx1	BiCS4	TLC	9804	4229	100.83	48.23
128GB	CL10	UHS-I (Grade 3)	V30	A2	PS8229	512Gbx2	BiCS4	TLC	9804	4577	100.91	92.73
256GB	CL10	UHS-I (Grade 3)	V30	A2	PS8229	512Gbx4	BiCS4	TLC	9804	4789	100.91	95.46
128GB	CL10	UHS-I (Grade 3)	V30	A2	PS8229	512Gbx2	BiCS5	TLC	9804	5137	100.85	95.14
256GB	CL10	UHS-I (Grade 3)	V30	A2	PS8229	512Gbx4	BiCS5	TLC	9804	5370	100.85	95.66
512GB	CL10	UHS-I (Grade 3)	V30	A2	PS8229	512Gbx8	BiCS5	TLC	9804	5348	100.85	95.59
128GB	CL10	UHS-I (Grade 3)	V30	A2	PS8229	1Tbx1	BiCS6	TLC	9804	4453	100.86	44.32

Capacity	Class	UHS-I	VSC	APP PERF Class	Controller	Flash			TestMetrix IOPs test		TestMetrix Performance	
						Density	Process	Bit- per- cell	Read (IOPs)	Write (IOPs)	Read (MB/s)	Write (MB/s)
256GB	CL10	UHS-I (Grade 3)	V30	A2	PS8229	1Tbx2	BiCS6	TLC	9804	4829	100.86	51.52
512GB	CL10	UHS-I (Grade 3)	V30	A2	PS8229	1Tbx4	BiCS6	TLC	9804	5117	100.85	56.83

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1. INTRODUCTION



1.1. General Description

The micro Secure Digital (microSD) card is fully compliant with the standards released by the SD Card Association. The Command List supports [Part 1 Physical Layer Specification Ver6.10 Final] definitions. Card capacities of the nonsecure area and secure area (if needed) support [Part 3 Security Specification Ver7.0] Specifications.

The microSD card comes with an 8-pin interface, designed to operate at a maximum frequency of 208MHz. It can alternate communication protocol between the SD mode and SPI mode. It performs data error detection and correction with very low power consumption. microSD card are one of the most popular removable storage cards today due to its high performance, good reliability and wide compatibility.

1.2. Flash Management

1.2.1. Error Correction Code

Flash memory cells will deteriorate with use, which might generate random bit errors in the stored data. Thus, microSD card applies ECC Algorithm, which can detect and correct errors during Read processes, ensuring data is read correctly, as well as protecting data from corruption.

1.2.2. Wear Leveling

NAND Flash devices can only undergo a limited number of program/erase cycles, and in most cases, the flash media are not used evenly. If some area gets updated more frequently than others, the lifetime of the device would be reduced significantly. Thus, Wear Leveling technique is applied to extend the lifespan of NAND Flash by evenly distributing write and erase cycles across the media.

Phison provides advanced Wear Leveling algorithm, which can efficiently spread out the flash usage through the whole flash media area. Moreover, by implementing both dynamic and static Wear Leveling algorithms, the life expectancy of the NAND Flash is greatly improved.

1.2.3. Bad Block Management

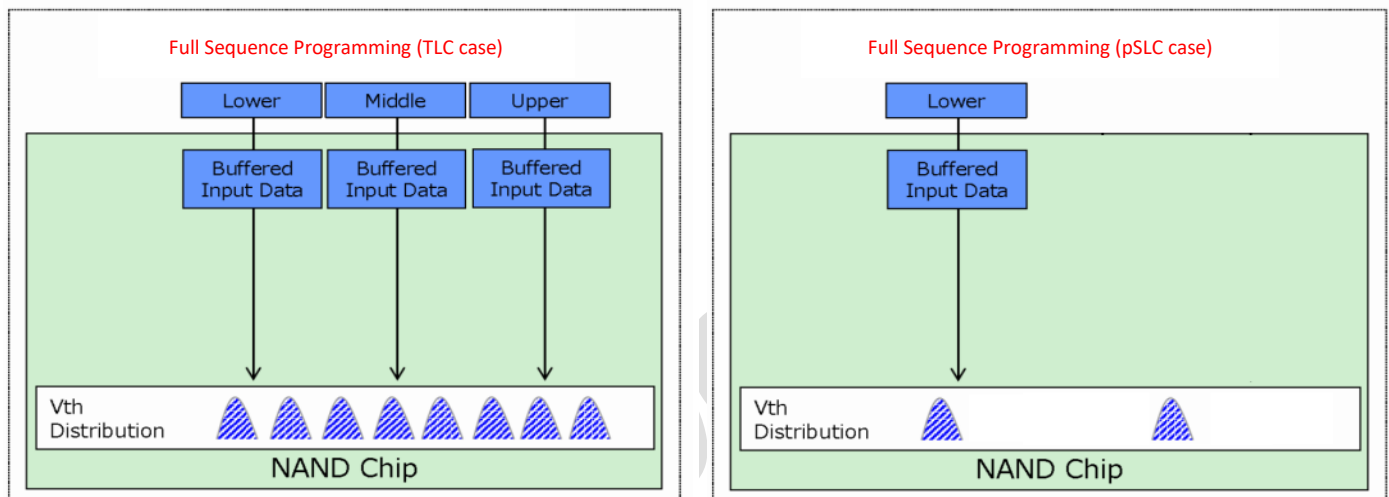
Bad blocks are blocks that include one or more invalid bits, and their reliability is not guaranteed. Blocks that are identified and marked as bad by the manufacturer are referred to as “Initial Bad Blocks”. Bad blocks that are developed during the lifespan of the flash are named “Later Bad Blocks”. Phison implements an efficient

bad block management algorithm to detect the factory-produced bad blocks and manages any bad blocks that appear with use. This practice further prevents data being stored into bad blocks and improves the data reliability.

1.2.4. Pseudo SLC (for pSLC product only)

Pseudo SLC can be considered as an extended version of TLC. While TLC does Full Sequence Programming into 8 Vth distribution, pSLC only does Lower page programming into 2 Vth distribution. Accordingly, because only Lower pages are programmed, pSLC provides better performance and endurance than TLC. Moreover, pSLC performs similarly with SLC, yet pSLC is more cost-effective.

Figure 1-1 Cell Content of TLC (Left) and Pseudo SLC (Right)



2. PRODUCT SPECIFICATIONS



- **Compliant Specifications - SD Memory Card Specifications:**
 - Compliant with Part 1 Physical Layer Specification Ver. 6.10
 - Compliant with Part 2 File System Specification Ver. 3.00
 - Compliant with Part 3 Security Specification Ver. 7.00
 - microSD Card Addendum Ver4.20
 - **Support SD SPI mode**
 - **Bus Speed Mode (use 4 parallel data lines)**
 - **Non-UHS Mode**
 - Default speed mode: 3.3V signaling, frequency up to 25MHz, up to 12.5 MB/sec
 - High speed mode: 3.3V signaling, frequency up to 50MHz, up to 25 MB/sec
 - **UHS Mode**
 - SDR12: SDR up to 25MHz, 1.8V signaling
 - SDR25: SDR up to 50MHz, 1.8V signaling
 - SDR50: 1.8V signaling, frequency up to 100MHz, up to 50 MB/sec
 - SDR104: 1.8V signaling, frequency up to 208MHz, up to 104MB/sec
 - DDR50: 1.8V signaling, frequency up to 50MHz, sampled on both clock edges, up to 50 MB/sec
- NOTES:** 1. Timing in 1.8V signaling is different from that of 3.3V signaling.
2. To properly run the UHS mode, please ensure the device supports UHS-I mode.
- **The command list supports [Part 1 Physical Layer Specification Ver. 6.10] definitions**
 - Command list are described in “Table 3-2 SD mode Command Set ” and “Table 3-3 SPI mode Command Set” in this document
 - **Copyrights Protection Mechanism**
 - Compliant with Part 1 Physical Layer Specification ver. 6.10, CPRM is Optional in microSDHC/microSDXC.
 - **Support Hot Plug**
 - Card removal during read operation will never harm the content
 - **Password Protection of cards (optional)**
 - **Designed for read intensive and write intensive cards**
 - **Built-in write protection features (permanent and temporary)**
 - **Write Protect feature using mechanical switch (Full SD Card only)**

- **Electrostatic Discharge(ESD)**
 - ESD protection in pads (contact discharge).
 - ESD protection in non-contact pad area (air discharge).
- **Operation voltage range: 2.7V ~ 3.6V**
- **Temperature Range**
 - Operation Temp. Range: -25°C~85°C
 - Storage Temp. Range: -40°C~85°C

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3. ELECTRICAL INTERFACE OUTLINES



3.1. microSD Card Pins

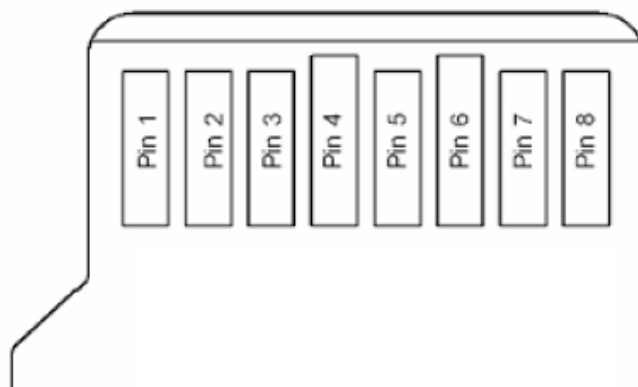


Figure 3-1 microSD Card Pin assignment (Back View of the card)

Table 3-1 microSD Memory Card Pad Assignment

pin	SD Mode			SPI Mode		
	Name	Type ¹	Description	Name	Type	Description
1	DAT2	I/O/PP	Data Line [bit2]	RSV		
2	CD/DAT3 ²	I/O/PP ³	Card Detect/ Data Line [bit3]	CS	I ³	Chip Select (neg. true)
3	CMD	PP	Command/Response	DI	I	Data In
4	V _{DD}	S	Supply voltage	V _{DD}	S	Supply voltage
5	CLK	I	Clock	SCLK	I	Clock
6	V _{SS}	S	Supply voltage ground	V _{SS}	S	Supply voltage ground
7	DAT0	I/O/PP	Data Line [bit0]	DO	O/PP	Data Out
8	DAT1	I/O/PP	Data Line [bit1]	RSV		

- (1) S: power supply, I: input; O: output using push-pull drivers; PP: I/O using push-pull drivers.
- (2) The extended DAT lines (DAT1-DAT3) are input on power up. They start to operate as DAT lines after SET_BUS_WIDTH command. The Host shall keep its own DAT1-DAT3 lines in input mode as well while they are not used. It is defined so in order to keep compatibility to MultiMedia Cards.
- (3) At power up, this line has a 50KOhm pull up enabled in the card. This resistor serves two functions: Card detection and Mode Selection. For Mode Selection, the host can drive the line high or let it be pulled high to select SD mode. If the host wants to select SPI mode, it should drive the line low. For Card detection, the host detects that the line is pulled high. This pull-up should be disconnected by the user during regular data transfer with SET_CLR_CARD_DETECT (ACMD42) command.

3.2. microSD Card Bus Topology

The microSD card supports 2 alternative communication protocols, SD and SPI BUS mode.

Host can choose either one of both bus mode, same data can be read or written by both modes. SD mode allows 4-bits data transfer way, it provides high performance. SPI mode supports 1-bit data transfer and of course the performance is lower compared to SD mode.

3.3. microSD Bus Mode Protocol

In default speed, the microSD Memory Card bus has a single master (application); multiple slaves (Cards), synchronous star topology (refer to Figure 3-2). In high speed and UHS-I, the microSD Memory Card bus has a single master (application) and single slave (card), synchronous point to point topology. Clock, power and ground signals are common to all cards. Command (CMD) and data (DAT0-DAT3) signals are dedicated to each card providing continues point to point connection to all the cards.

During initialization process commands are sent to each card individually, allowing the application to detect the cards and assign logical addresses to the physical slots. Data is always sent (received) to (from) each card individually. However, in order to simply the handling of the card stack, after the initialization process, all commands may be sent concurrently to all cards. Addressing information is provided in the command packet.

SD bus allows dynamic configuration of the number of data lines. After power up, by default, the microSD Memory Card will use only DAT0 for data transfer. After initialization the host can change the bus width (number of data active lines). This feature allows easy tradeoff between HW cost and system performance. Note that while DAT1 to DAT3 are not in use, the related Host's DAT lines should be in tri-state (input mode).

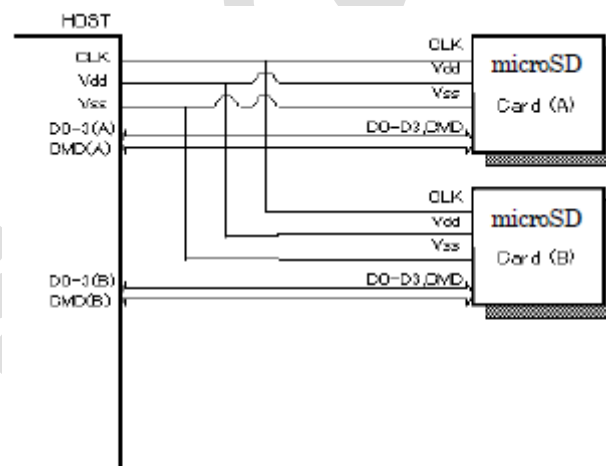


Figure 3-2 SD Memory Card System Bus Topology

The microSD bus includes the following signals:

CLK: Host to card clock signal

CMD: Bidirectional Command/Response signal

DAT0-DAT3: 4 Bidirectional data signals

VDD, VSS1, VSS2: Power and ground signals

Table 3-2 SD Mode Command Set

Card Command Class (CCC)	0 basic	1 Comm and Queue	2 block read	3 reserved	4 block write	5 erase	6 write protection	7 lock card	8 application specific	9 I/O mode	10 switch	11 extension
CMD0	+											
CMD2	+											
CMD3	+											
CMD4	+											
CMD5										+		
CMD6											+	
CMD7	+											
CMD8	+											
CMD9	+											
CMD10	+											
CMD11	+											
CMD12	+											
CMD13	+											
CMD15	+											
CMD16			+		+			+				
CMD17			+									
CMD18			+									
CMD19			+									
CMD20			+		+							
CMD21												+
CMD23			+		+							
CMD24					+							
CMD25					+							
CMD27					+							
CMD28							+					
CMD29							+					
CMD30							+					
CMD32						+						
CMD33						+						
CMD34-37											+	
CMD38						+						
CMD40								+				
CMD42								+				
CMD43-47		+										
CMD48												+
CMD49												+
CMD50											+	
CMD52										+		
CMD53										+		

Card Command Class (CCC)	0 basic	1 Comm and Queue	2 block read	3 reserved	4 block write	5 erase	6 write protection	7 lock card	8 application specific	9 I/O mode	10 switch	11 extension
CMD55									+			
CMD56									+			
CMD57											+	
CMD58												+
CMD59												+
ACMD6									+			
ACMD13									+			
ACMD14									+			
ACMD15									+			
ACMD16									+			
ACMD22									+			
ACMD23									+			
ACMD28									+			
ACMD41									+			
ACMD42									+			
ACMD51									+			

Commands	Support requirements
CMD0	Mandatory
CMD2	Mandatory
CMD3	Mandatory
CMD4	Mandatory
CMD5	Optional
CMD6	Mandatory for cards version 1.10 and after
CMD7	Mandatory
CMD8	Mandatory for cards version 2.00 and after
CMD9	Mandatory
CMD10	Mandatory
CMD11	Mandatory for cards supporting UHS-I. Optional for cards that do not support UHS-I.
CMD12	Mandatory
CMD13	Mandatory
CMD15	Mandatory
CMD16	Mandatory
CMD17	Mandatory
CMD18	Mandatory
CMD19	Mandatory for cards supporting UHS-I. Optional for cards that do not support UHS-I.

Commands	Support requirements
CMD20	Not supported for SDSC cards. Mandatory for SDHC and SDXC cards that support Video Speed Class. Optional for SDHC cards that support: a) Speed Class; or b) UHS Speed Grade, and do not support Video Speed Class. Mandatory for SDXC cards that support Speed Class or UHS Speed Grade.
CMD21	Optional
CMD23	Not supported for SDSC cards. Mandatory for SDHC and SDXC cards that support UHS104. Optional for SDHC and SDXC cards that do not support UHS104.
CMD24	Mandatory for writable types of cards
CMD25	Mandatory for writable types of cards
CMD27	Mandatory for writable types of cards
CMD28	Optional
CMD29	Optional
CMD30	Optional
CMD32	Mandatory for writable types of cards
CMD33	Mandatory for writable types of cards
CMD34-37	Optional for cards version 1.10 and after
CMD38	Mandatory for writable types of cards Discard and FULE support is optional
CMD40	Optional
CMD42	Optional for cards version 1.01 and 1.10. Mandatory for cards version 2.00 and after. COP support is optional for CMD42
CMD43-47	Mandatory for cards supporting Command Queue
CMD48	Optional Mandatory for cards supporting Performance Enhancement functions (refer to 5.8.2)
CMD49	Optional Mandatory for cards supporting Performance Enhancement functions (refer to 5.8.2)
CMD50	Optional for cards version 1.10 and after
CMD52	Optional
CMD53	Optional
CMD55	Mandatory
CMD56	Mandatory
CMD57	Optional for cards version 1.10 and after
CMD58	Optional
CMD59	Optional
ACMD6	Mandatory
ACMD13	Mandatory
ACMD14	Optional
ACMD15	Optional
Commands	Support requirements
ACMD16	Optional
ACMD22	Mandatory for writable types of cards
ACMD23	Mandatory for writable types of cards
ACMD28	Optional
ACMD41	Mandatory
ACMD42	Mandatory
ACMD51	Mandatory

3.4. SPI Bus Mode Protocol

While the SD Memory Card channel is based on command and data bit streams that are initiated by a start bit and terminated by a stop bit, the SPI channel is byte oriented. Every command or data block is built for 8-bit bytes and is byte aligned with the CS signal (i.e. the length is a multiple of 8 clock cycles). The card starts to count SPI bus clock cycle at the assertion of the CS signal. Every command or data token shall be aligned with 8-clock cycle boundary.

Similar to the SD Memory Card Protocol, the SPI messages consist of command, response and data-block tokens.

The advantage of SPI mode is reducing the host design effort, especially for MMC host side, it just be modified by little change. Note: please use SD card specification to implement SPI mode function, not use MMC specification. For example, SPI mode is initialized by ACMD41, and the registers are different from MMC card, especially CSD register.

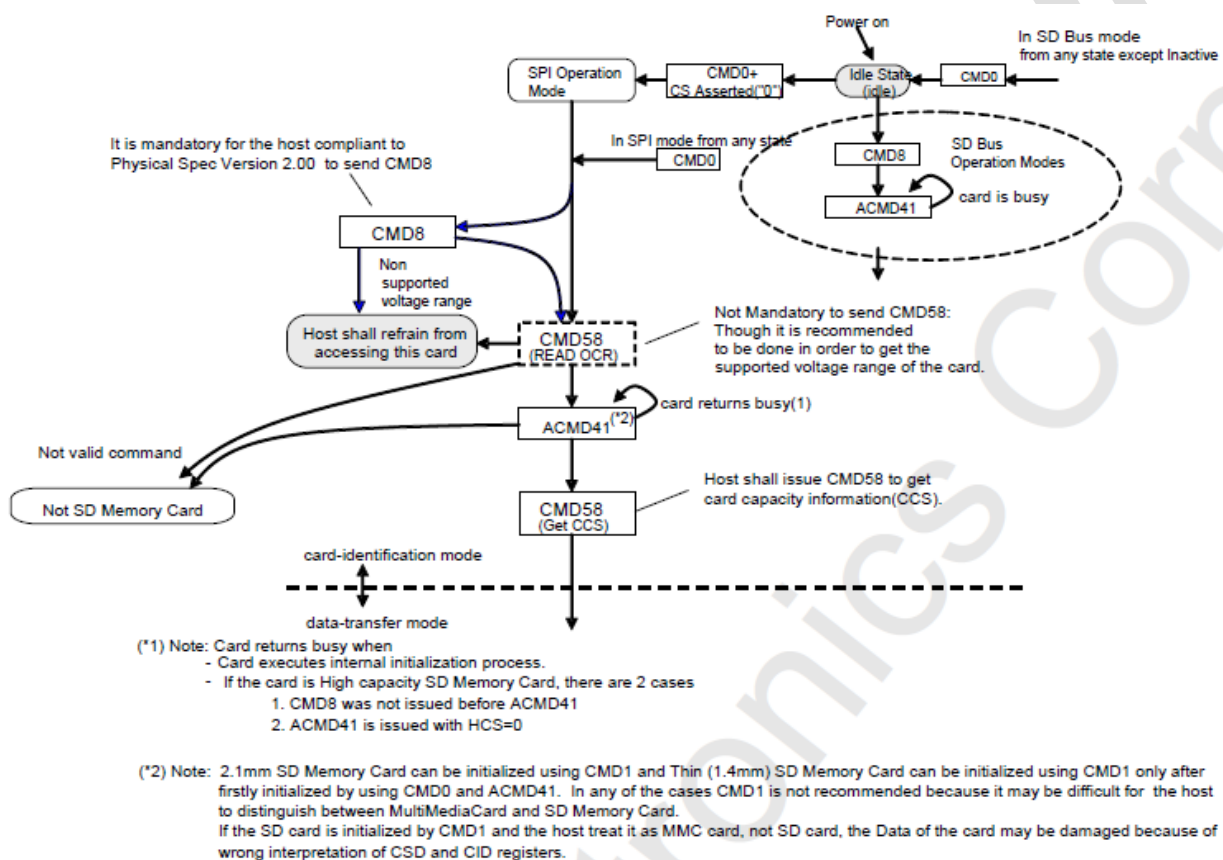


Figure 3-3 SD Memory Card State Diagram (SPI mode)

Table 3-3 SPI Mode Command Set

Card Command Class (CCC)		0	1	2	3	4	5	6	7	8	9	10	11
Supported commands	class description	basic	reser ved	block read	reser ved	block write	erase	write pro-tection	lock card	appli- cat- ion spe- cific	I/O mode	switch	reser ved
CMD0	Mandatory	+											
CMD1	Mandatory	+											
CMD5	Optional										+		
CMD6 ²	Mandatory											+	
CMD8 ³	Mandatory	+											
CMD9	Mandatory	+											
CMD10	Mandatory	+											
CMD12	Mandatory	+											
CMD13	Mandatory	+											
CMD16	Mandatory			+		+			+				
CMD17	Mandatory			+									
CMD18	Mandatory			+									
CMD24	Mandatory ¹					+							
CMD25	Mandatory ¹					+							

Card Command Class (CCC)		0	1	2	3	4	5	6	7	8	9	10	11
Supported commands	class description	basic	reser ved	block read	reser ved	block write	erase	write pro-tection	lock card	appli- cat- ion spe- cific	I/O mode	switch	reser ved
CMD27	Mandatory ¹					+							
CMD28	Optional							+					
CMD29	Optional							+					
CMD30	Optional							+					
CMD32	Mandatory ¹						+						
CMD33	Mandatory ¹						+						
CMD34-37 ²	Optional											+	
CMD38	Mandatory ¹						+						
CMD42 ⁴	(Note 4)								+				
CMD50 ²	Optional											+	
CMD52	Optional										+		
CMD53	Optional										+		
CMD55	Mandatory									+			
CMD56	Mandatory									+			
CMD57 ²	Optional											+	
CMD58	Mandatory	+											
CMD59	Mandatory	+											
ACMD13	Mandatory									+			
ACMD22	Mandatory ¹									+			
ACMD23	Mandatory ¹									+			
ACMD41	Mandatory									+			
ACMD42	Mandatory									+			
ACMD51	Mandatory									+			

Note (1): The commands related write and erase are mandatory only for the Writable types of Cards.

Note (2): This command was defined in spec version 1.10

Note (3): This command is newly defined in version 2.00

Note (4): This command is optional in Version 1.01 and 1.10 and mandatory from Version 2.00. COP support is optional for CMD42

3.5. microSD card initialization

Figure 3-4 presents the initialization flow chart for UHS-I hosts and Figure 3-5 shows sequence of commands to perform voltage switch.

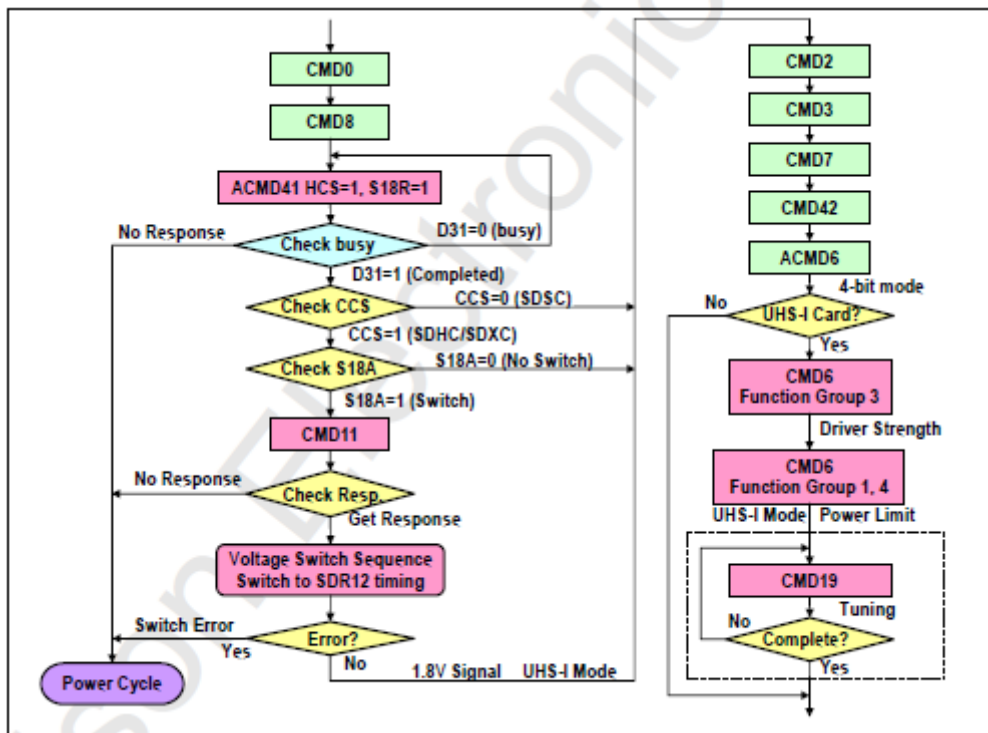


Figure 3-4 UHS-I Host Initialization Flow Chart

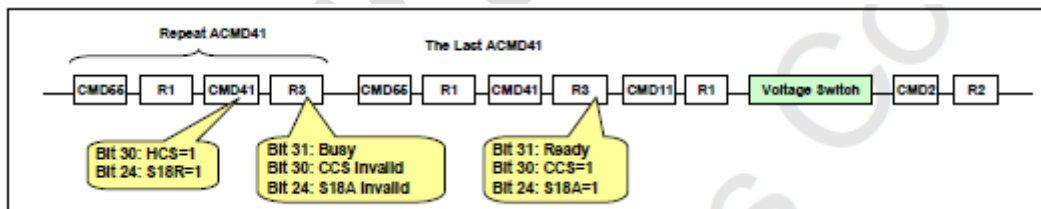


Figure 3-5 ACMD41 Timing Followed by Voltage Switch Sequence

When signaling level is 3.3V, host repeats to issue ACMD41 with HCS=1 and S18R=1 until the response indicates ready. The argument (HCS and S18R) of the first ACMD41 is effective but the all following ACMD41 should be issued with the same argument.

If Bit31 indicates ready, host needs to check CCS and S18A.

The card indicates S18A=0, which means that voltage switch is not allowed and the host needs to use current signaling level.

Table 3-4 S18R and S18A Combinations

Current Signaling Level	S18R	S18A	Comment
3.3V	0	0	1.8V signaling is not requested
	1	0	The card does not support 1.8V signaling
	1	1	Start signal voltage switch sequence
1.8V	X	0	Already switched to 1.8V

To change signaling level at the same time between host and card, signal voltage switch sequence is invoked by CMD11 as shown in Figure 3-6. CMD11 is issued only when S18A=1 in the response of ACMD41.

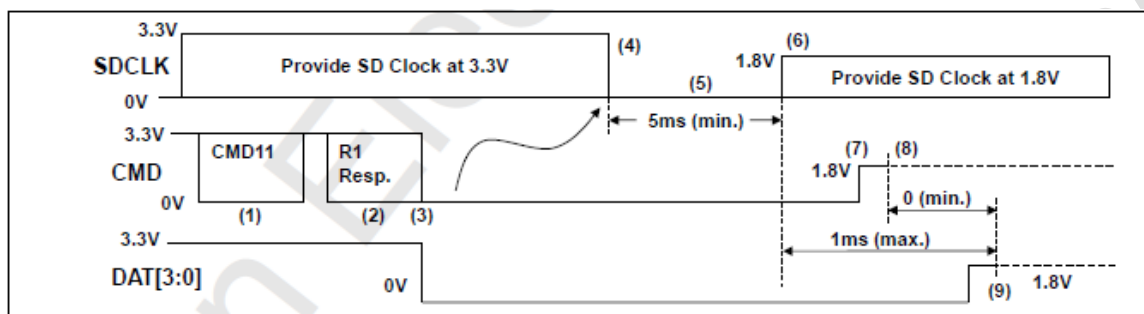


Figure 3-6 Signal Voltage Switch Sequence

4. ENVIRONMENTAL SPECIFICATIONS



4.1. Environmental Conditions

Temperature and Humidity

- Temperature Range
 - Operational: -25°C ~ 85°C
 - Storage: -40°C ~ 85°C
- Humidity
 - Operational: RH = 95% under 25°C

Table 4-1 High-Temperature Test Condition

	Temperature	Humidity	Test Time
Operation	85°C	0% RH	96 hours
Storage	85°C	0% RH	500 hours

Result: No any abnormality is detected.

Table 4-2 Low-Temperature Test Condition

	Temperature	Humidity	Test Time
Operation	-25°C	0% RH	96 hours
Storage	-40°C	0% RH	168 hours

Result: No any abnormality is detected.

Table 4-3 High Humidity Test Condition

	Temperature	Humidity	Test Time
Operation	25°C	95% RH	1 hour
Storage	40°C	93% RH	500 hours

Result: No any abnormality is detected.

Shock

Table 4-4 Shock Specification

	Acceleration Force	Half Sin Pulse Duration
microSD card	500G	0.5ms

Result: No any abnormality is detected when power on.

Vibration

Table 4-5 Vibration Specification

	Condition		Vibration Orientation
	Frequency/Displacement	Frequency/Acceleration	
microSD card	20Hz~80Hz/1.52mm	80Hz~2000Hz/20G	Direction: X, Y, Z axis Duration: 30 min/direction

Result: No any abnormality is detected when power on.

Drop

Table 4-6 Drop Specification

	Height of Drop	Number of Drops
microSD card	150cm free fall	Direction: 6 face; 1 time/face

Result: No any abnormality is detected when power on.

Bending

Table 4-7 Bending Specification

	Force	Action
microSD card	≥ 10N	Hold for 1min; total 5 times.

Result: No any abnormality is detected when power on.

Torque

Table 4-8 Torque Specification

	Force	Action
microSD card	0.1N-m or ±2.5 deg	Hold 30 second/direction Total 5 cycles

Result: No any abnormality is detected when power on.

Durability Mating Cycle Test

Table 4-9 Mating Cycle Test Specification

	Number of Mating Cycle
microSD card	10,000 cycles

Result: No any abnormality is detected when power on.

Electrostatic Discharge (ESD)

Table 4-10 ESD Specification

Condition		Result
microSD card	Non-operating Contact: $\pm 4\text{KV}$; 5 times/Pin Air: $\pm 15\text{KV}$; 5 times/Position	PASS
	Operating Air: $\pm 8\text{KV}$; 10 times/Position (EN55024-61000-4-2)	B grade, PASS

Result: No any abnormality is detected when power on.

EMI Compliance

- FCC: CISPR22
- CE: EN55032
- BSMI 13438

5. SD CARD COMPARISON



Table 5-1 Comparing microSDSC, microSDHC, and microSDXC

	microSDSC	microSDHC	microSDXC
File System	FAT 12/16	FAT32	exFAT
Addressing Mode	Byte (1 byte unit)	Block (512 byte unit)	Block (512 byte unit)
HCS/CCS bits of ACMD41	Support	Support	Support
CMD8 (SEND_IF_COND)	Support	Support	Support
CMD16 (SET_BLOCKLEN)	Support	Support (Only CMD42)	Support (Only CMD42)
Partial Read	Support	Not Support	Not Support
Lock/Unlock Function	Mandatory	Mandatory	Mandatory
Write Protect Groups	Optional	Not Support	Not Support
Supply Voltage 2.7v – 3.6v (for operation)	Support	Support	Support
Total Bus Capacitance for each signal line	40pF	40pF	40pF
CSD Version (CSD_STRUCTURE Value)	1.0 (0x0)	2.0 (0x1)	2.0 (0x1)
Speed Class	Optional	Mandatory (Class 2 / 4 / 6 / 10)	Mandatory (Class 2 / 4 / 6 / 10)

Table 5-2 Comparing UHS Speed Grade Symbols

	U1 (UHS Speed Grade 1)	U3 (UHS Speed Grade 3)
Operable Under	*UHS-I Bus I/F, UHS-II Bus I/F	
SD Memory Card	microSDHC UHS-I and UHS-II, microSDXC UHS-I and UHS-II	
Mark		
Performance	10 MB/s minimum write speed	30 MB/s minimum write speed
Applications	Full higher potential of recording real-time broadcasts and capturing large-size HD videos.	Capable of recording 4K2K video.

*UHS (Ultra High Speed), the fastest performance category available today, defines bus-interface speeds up to 312 Megabytes per second for greater device performance. It is available on microSDXC and microSDHC memory cards and devices.

6. ELECTRICAL SPECIFICATIONS



6.1. Power Consumption

The table below is the power consumption of microSD card with different flash memory types.

Table 6-1 Power Consumption of microSD card

Flash Mode		Max. Power Up Current (uA)	Max. Standby Current (uA)	Max. Read Current (mA)	Max. Write Current (mA)
Default Speed Mode		250	1000	150 @ 3.6V	150 @ 3.6V
High Speed Mode		250	1000	200 @ 3.6V	200 @ 3.6V
UHS-I Mode	UHS50/DDR50	250	1000	400 @ 3.6V	400 @ 3.6V
	UHS104/ DDR50	250	1000	400 @ 3.6V	400 @ 3.6V

NOTES:

1. Power consumptions are measured at room temperature.

6.2. Working Rating

Item	Symbol	Parameter	MIN	MAX	Unit
1	T_a	Operating Temperature	-25	+85	°C
2	T_{st}	Storage Temperature	-40	+85	°C

Parameter	Symbol	Min	MAX	Unit
Operating Temperature	T_a	-25	+85	°C
V_{DD} Voltage	V_{DD}	2.7	3.6	V

6.3. DC Characteristic

6.3.1. Bus Operation Conditions for 3.3V Signaling

Table 6-2 Threshold Level for High Voltage Range

Parameter	Symbol	Min.	Max	Unit	Condition
Supply Voltage	V_{DD}	2.7	3.6	V	
Output High Voltage	V_{OH}	$0.75 \cdot V_{DD}$		V	$I_{OH} = -2mA$ V_{DD} Min
Output Low Voltage	V_{OL}		$0.125 \cdot V_{DD}$	V	$I_{OL} = 2mA$ V_{DD} Min
Input High Voltage	V_{IH}	$0.625 \cdot V_{DD}$	$V_{DD} + 0.3$	V	
Input Low Voltage	V_{IL}	$V_{SS} - 0.3$	$0.25 \cdot V_{DD}$	V	
Power Up Time			250	ms	From 0V to V_{DD} min

Table 6-3 Peak Voltage and Leakage Current

Parameter	Symbol	Min	Max.	Unit	Remarks
Peak voltage on all lines		-0.3	$V_{DD} + 0.3$	V	
All Inputs					
Input Leakage Current		-10	10	uA	
All Outputs					
Output Leakage Current		-10	10	uA	

Table 6-4 Threshold Level for 1.8V Signaling

Parameter	Symbol	Min.	Max	Unit	Condition
Supply Voltage	V_{DD}	2.7	3.6	V	
Regulator Voltage	V_{DDIO}	1.7	1.95	V	Generated by V_{DD}
Output High Voltage	V_{OH}	1.4	-	V	$I_{OH}=-2mA$
Output Low Voltage	V_{OL}	-	0.45	V	$I_{OL}=2mA$
Input High Voltage	V_{IH}	1.27	2.00	V	
Input Low Voltage	V_{IL}	$V_{SS}-0.3$	0.58	V	

Table 6-5 Input Leakage Current for 1.8V Signaling

Parameter	Symbol	Min	Max.	Unit	Remarks
Input Leakage Current		-2	2	μA	DAT3 pull-up is disconnected.

6.3.2. Bus Signal Line Load

Bus Operation Conditions – Signal Line's Load

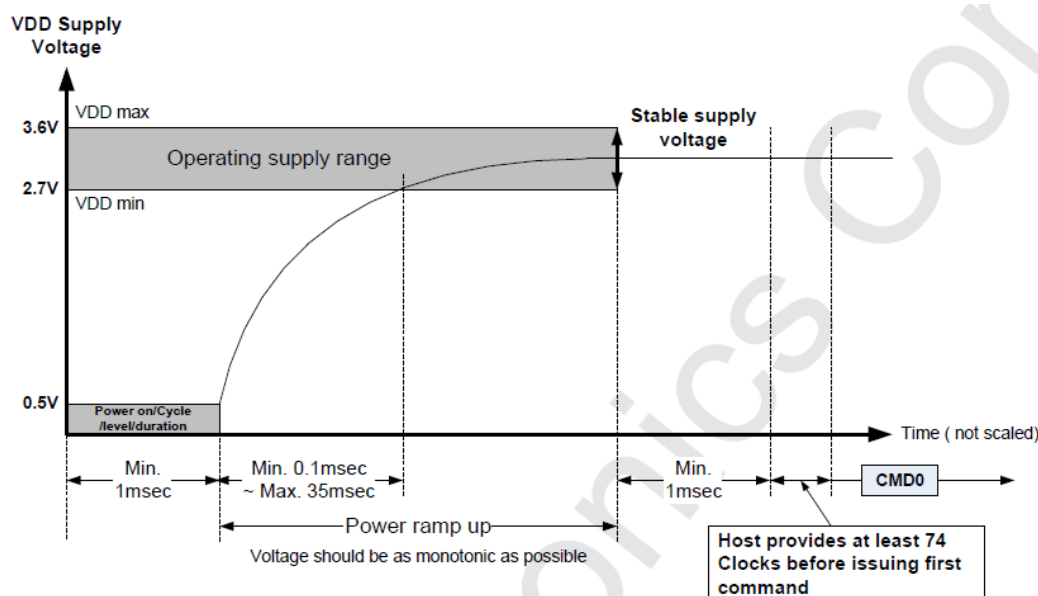
Total Bus Capacitance = $C_{HOST} + C_{BUS} + N C_{CARD}$

Parameter	symbol	Min	Max	Unit	Remark
Pull-up resistance	R_{CMD} R_{DAT}	10	100	$k\Omega$	to prevent bus floating
Total bus capacitance for each signal line	C_L		40	pF	1 card $C_{HOST}+C_{BUS}$ shall not exceed 30 pF
Card Capacitance for each signal pin	C_{CARD}		10^1	pF	
Maximum signal line inductance			16	nH	
Pull-up resistance inside card (pin1)	R_{DAT3}	10	90	$k\Omega$	May be used for card detection
Capacity Connected to Power Line	C_C		5	μF	To prevent inrush current

<Note 1> PS8210 is SD and eMMC(4.51) controller, so the maximum of eMMC capacitance will be 12pF.

6.3.3. Power Up Time of Host

The host needs to keep power line level less than 0.5V and more than 1ms before power ramp up.



Power On or Power Cycle

Followings are requirements for Power on and Power cycle to assure a reliable microSD Card hard reset.

- (1) Voltage level shall be below 0.5V.
- (2) Duration shall be at least 1ms.

Power Supply Ramp Up

The power ramp up time is defined from 0.5V threshold level up to the operating supply voltage which is stable between VDD (min.) and VDD (max.) and host can supply SDCLK.

Followings are recommendations of Power ramp up:

- (1) The voltage of power ramp up should be monotonic as much as possible.
- (2) The minimum ramp up time should be 0.1ms.
- (3) The maximum ramp up time should be 35ms for 2.7-3.6V power supply.
- (4) Host shall wait until VDD is stable.
- (5) After 1ms VDD stable time, the host provides at least 74 clocks before issuing the first command.

Power Down and Power Cycle

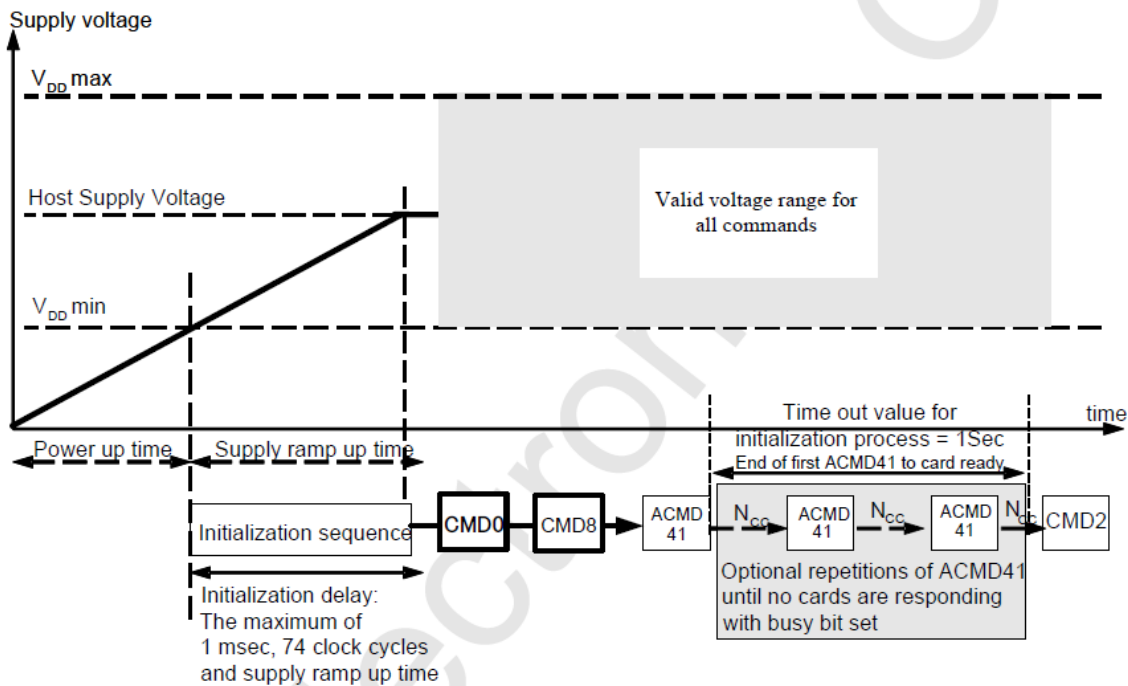
- (1) When the host shuts down the power, the card V_{DD} shall be lowered to less than 0.5V for a minimum period of 1ms. During power down, DAT, CMD, and CLK should be disconnected or driven to logical 0 by the host to avoid a situation that the operating current is drawn through the signal lines.
- (2) If the host needs to change the operating voltage, a power cycle is required. Power cycle means the power is turned off and supplied again. A power cycle is also needed for accessing cards that are

already in *Inactive State*. To create a power cycle the host shall follow the power down description before power up the card (i.e. the card V_{DD} shall be once lowered to less than 0.5Volt for a minimum period of 1ms).

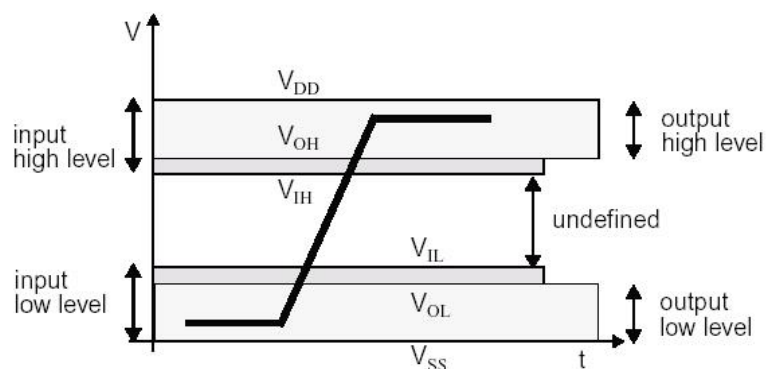
6.3.4. Power Up Time of Card

A device shall be ready to accept the first command within 1ms from detecting V_{DD} min.

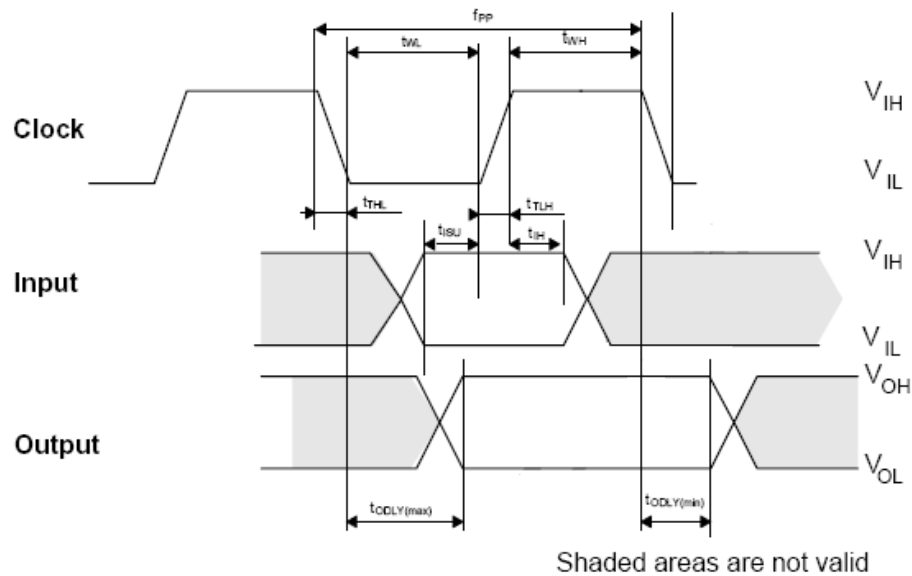
The device may use up to 74 clocks for preparation before receiving the first command.



6.4. AC Characteristic



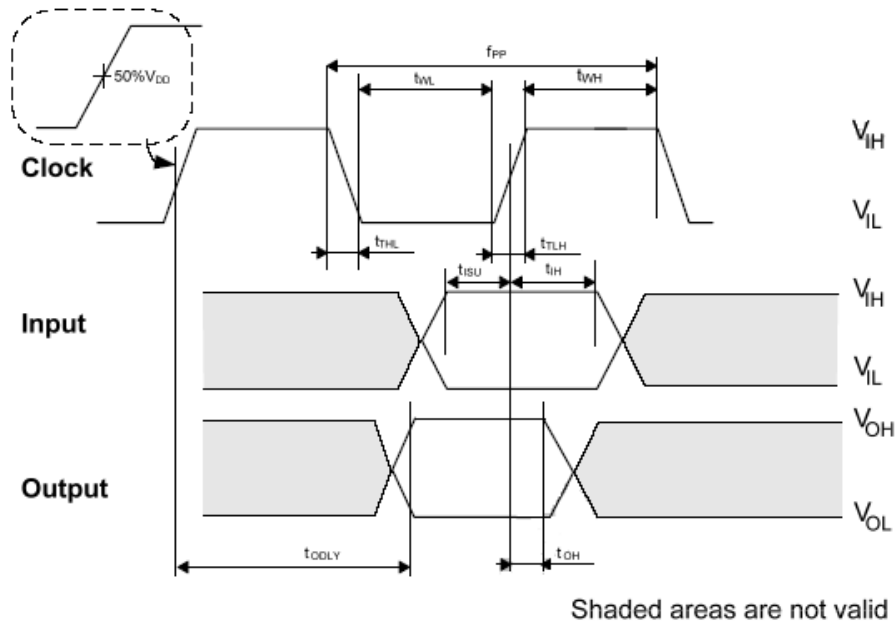
6.4.1. microSD Interface Timing (Default)



Parameter	Symbol	Min	Max	Unit	Remark
Clock CLK (All values are referred to min(V_{IH}) and max(V_{IL}))					
Clock frequency Data Transfer Mode	f _{PP}	0	25	MHz	C _{card} ≤ 10 pF (1 card)
Clock frequency Identification Mode	f _{OD}	0 ₍₁₎ /100	400	kHz	C _{card} ≤ 10 pF (1 card)
Clock low time	t _{WL}	10		ns	C _{card} ≤ 10 pF (1 card)
Clock high time	t _{WH}	10		ns	C _{card} ≤ 10 pF (1 card)
Clock rise time	t _{TLH}		10	ns	C _{card} ≤ 10 pF (1 card)
Clock fall time	t _{THL}		10	ns	C _{card} ≤ 10 pF (1 card)
Inputs CMD, DAT (referenced to CLK)					
Input set-up time	t _{ISU}	5		ns	C _{card} ≤ 10 pF (1 card)
Input hold time	t _{IH}	5		ns	C _{card} ≤ 10 pF (1 card)
Outputs CMD, DAT (referenced to CLK)					
Output Delay time during Data Transfer Mode	t _{ODLY}	0	14	ns	C _L ≤ 40 pF (1 card)
Output Delay time during Identification Mode	t _{ODLY}	0	50	ns	C _L ≤ 40 pF (1 card)

(1) 0Hz means to stop the clock. The given minimum frequency range is for cases where continuous clock is required.

6.4.2. microSD Interface Timing (High-Speed Mode)



Parameter	Symbol	Min	Max	Unit	Remark
Clock CLK (All values are referred to min(V_{IH}) and max(V_{IL}))					
Clock frequency Data Transfer Mode	f _{PP}	0	50	MHz	C _{card} ≤ 10 pF (1 card)
Clock low time	t _{WL}	7		ns	C _{card} ≤ 10 pF (1 card)
Clock high time	t _{WH}	7		ns	C _{card} ≤ 10 pF (1 card)
Clock rise time	t _{TLH}		3	ns	C _{card} ≤ 10 pF (1 card)
Clock fall time	t _{THL}		3	ns	C _{card} ≤ 10 pF (1 card)
Inputs CMD, DAT (referenced to CLK)					
Input set-up time	t _{ISU}	6		ns	C _{card} ≤ 10 pF (1 card)
Input hold time	t _{IH}	2		ns	C _{card} ≤ 10 pF (1 card)
Outputs CMD, DAT (referenced to CLK)					
Output Delay time during Data Transfer Mode	t _{ODLY}		14	ns	C _L ≤ 40 pF (1 card)
Output Hold time	T _{OH}	2.5		ns	C _L ≤ 15 pF (1 card)
Total System capacitance of each line ¹	C _L		40	pF	C _L ≤ 15 pF (1 card)

(1) In order to satisfy severe timing, the host shall drive only one card.

6.4.3. microSD Interface Timing (SDR12, SDR25, SDR50 and SDR104 Modes)

Input

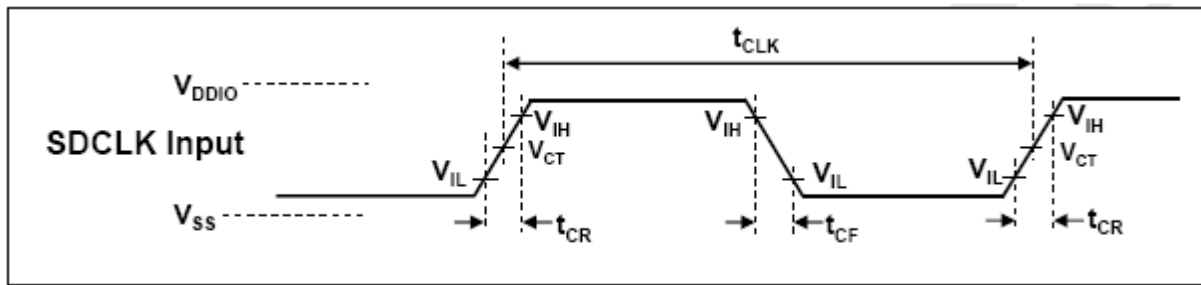
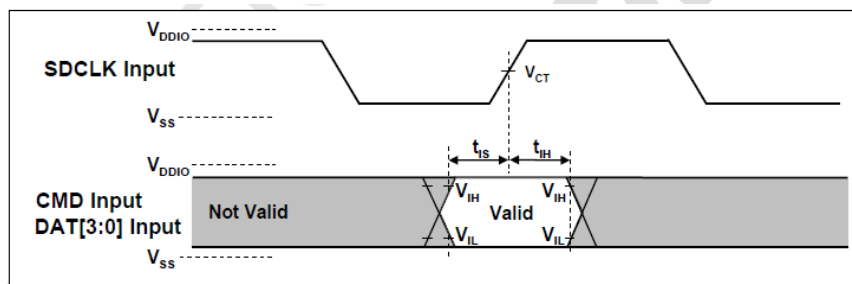


Table 6-6 Clock Signal Timing

Symbol	Min	Max	Unit	Remark
t_{CLK}	4.80	-	ns	208MHz (Max.), Between rising edge, $V_{CT} = 0.975V$
t_{CR}, t_{CF}	-	$0.2 * t_{CLK}$	ns	$t_{CR}, t_{CF} < 0.96ns$ (max.) at 208MHz, $C_{CARD} = 10pF$ $t_{CR}, t_{CF} < 2.00ns$ (max.) at 100MHz, $C_{CARD} = 10pF$ The maximum value of t_{CR}, t_{CF} is 10ns regardless of clock frequency
Clock Duty	30	70	%	

SDR50 and SDR104 Input Timing

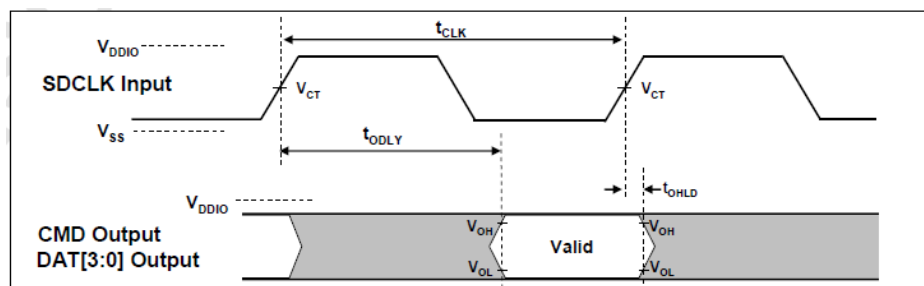


Card Input Timing

Symbol	Min	Max	Unit	SDR104 Mode
t_{IS}	1.40	-	ns	$C_{CARD} = 10pF, V_{CT} = 0.975V$
t_{IH}	0.80^1	-	ns	$C_{CARD} = 5pF, V_{CT} = 0.975V$
Symbol	Min	Max	Unit	SDR50 Mode
t_{IS}	3.00	-	ns	$C_{CARD} = 10pF, V_{CT} = 0.975V$
t_{IH}	0.80^1	-	ns	$C_{CARD} = 5pF, V_{CT} = 0.975V$

<Note 1> PS8210 is SD and eMMC(4.51) controller, so the maximum C_{CARD} becomes 12pF and minimum of t_{IH} will be 1.10 ns.

Output(SDR12, SDR25, SDR50)



Output Timing of Fixed Data Window

Table 6-7 Output Timing of Fixed Data Window (SDR12, SDR25, SDR50)

Symbol	Min	Max	Unit	Remark
t_{ODLY}	-	7.5	ns	$t_{CLK} \geq 10.0ns$, $C_L = 30pF$, using driver Type B, for SDR50
t_{ODLY}	-	14	ns	$t_{CLK} \geq 20.0ns$, $C_L = 40pF$, using driver Type B, for SDR25 and SDR12,
T_{OH}	1.5	-	ns	Hold time at the t_{ODLY} (min.), $C_L = 15pF$

Output(SDR104 Modes)

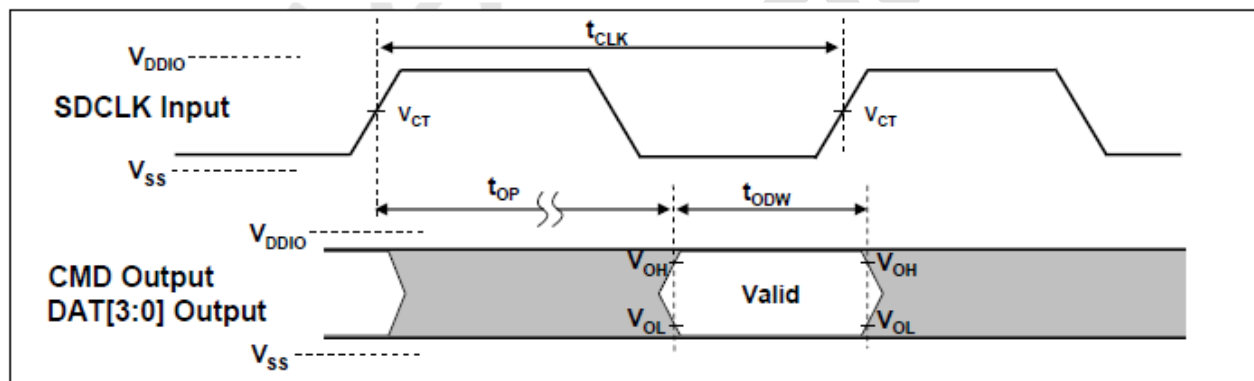
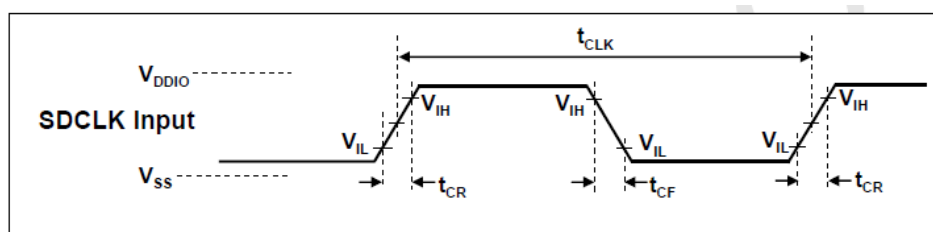


Table 6-8 Output Timing of Variable Window (SDR104)

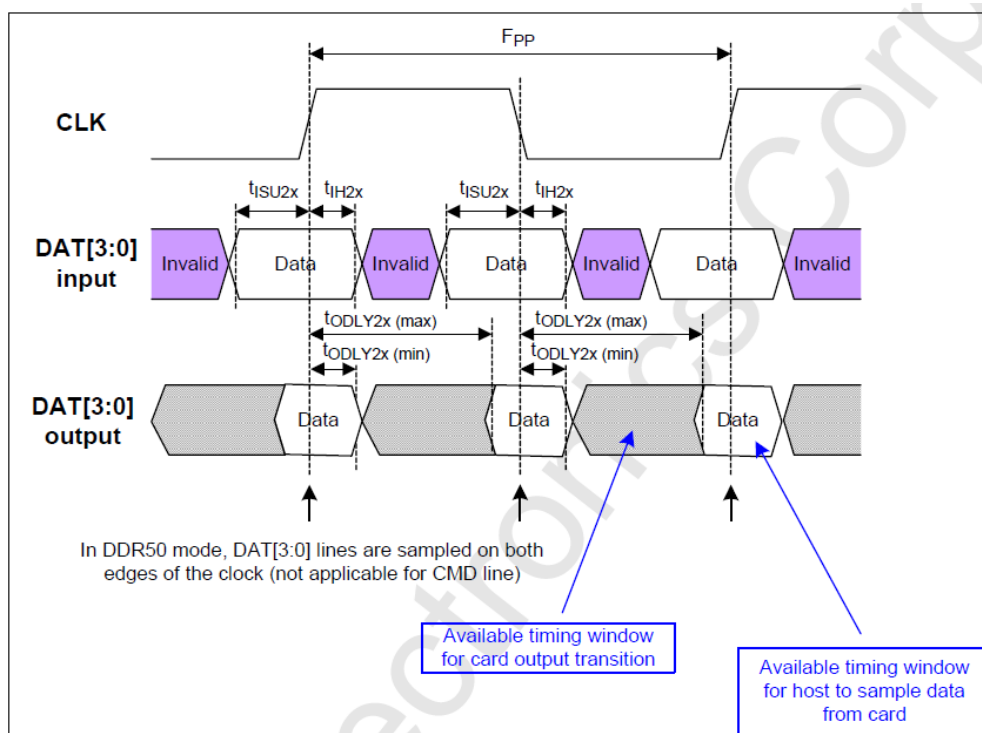
Symbol	Min	Max	Unit	Remark
t_{OP}	0	2	UI	Card Output Phase
Δt_{OP}	-350	+1550	ps	Delay variable due to temperature change after tuning
t_{ODW}	0.60	-	UI	$t_{ODW} = 2.88ns$ at 208MHz

6.4.4. microSD Interface Timing (DDR50 Mode)



Clock Signal Timing

Symbol	Min	Max	Unit	Remark
t_{CLK}	20	-	ns	50MHz (Max.), Between rising edge
t_{CR}, t_{CF}	-	$0.2 * t_{CLK}$	ns	$t_{CR}, t_{CF} < 4.00ns$ (max.) at 50MHz, $C_{CARD}=10pF$
Clock Duty	45	55	%	



Timing Diagram DAT Inputs/Outputs Referenced to CLK in DDR50 Mode

Table 6-9 Bus Timings – Parameters Values (DDR50 Mode)

Parameter	Symbol	Min	Max	Unit	Remark
Input CMD (referenced to CLK rising edge)					
Input set-up time	t_{ISU}	3	-	ns	$C_{card} \leq 10 pF$ (1 card)
Input hold time	t_{IH}	0.8	-	ns	$C_{card} \leq 10 pF$ (1 card)
Output CMD (referenced to CLK rising edge)					
Output Delay time during Data Transfer Mode	t_{ODLY}		13.7	ns	$C_L \leq 30 pF$ (1 card)
Output Hold time	T_{OH}	1.5	-	ns	$C_L \geq 15 pF$ (1 card)
Inputs DAT (referenced to CLK rising and falling edges)					
Input set-up time	t_{ISU2x}	3	-	ns	$C_{card} \leq 10 pF$ (1 card)
Input hold time	t_{IH2x}	0.8	-	ns	$C_{card} \leq 10 pF$ (1 card)
Outputs DAT (referenced to CLK rising and falling edges)					
Output Delay time during Data Transfer Mode	t_{ODLY2x}	-	7.0	ns	$C_L \leq 25 pF$ (1 card)
Output Hold time	T_{OH2x}	1.5	-	ns	$C_L \geq 15 pF$ (1 card)

7. HOST SYSTEM DESIGN GUIDELINES



7.1. Efficient Data Writing to microSD Memory Card

In order to optimize sequential writing performance and WAF (Write Amplification Factor), it is recommended to use allocation unit (AU) writing.

It is recommended that Multiple_Block_Write shall be used as a command for writing data, and the size of data written by each command should be the FAT cluster x n (n: integer)

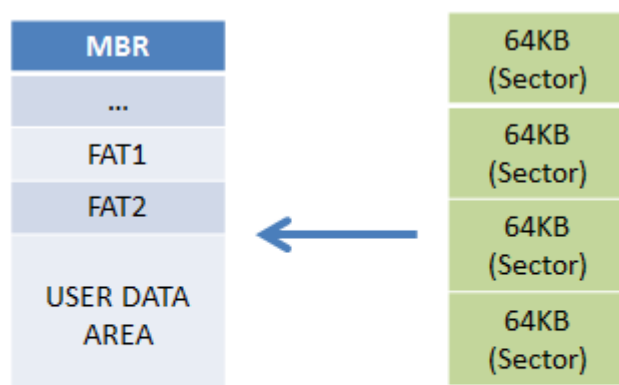
7.1.1. Write_Single_Block and Write_Multiple_Block

Write single block (CMD24) was written by one sector (512Bytes), which is suitable to write small area such like updating file system area (FAT). Besides, Write multiple blocks (CMD25) is a command for writing data to blocks that have sequential address per command, which is suitable to write large area such as user data.

Write multiple blocks with a cluster unit (512Byte x 128 Sectors = 64KByte) in the file system is an efficient access to the flash memory, it is obviously to provide higher speed to compared to single write block.

And it could be estimated that microSD card internal process would be reduced to save power consumption and flash write amplification factor, that is why the efficient data writing was recommended. To avoid the command issued by 512Bytes with single write block, software processes in the host device become faster.

For this operation, check the sectors in the microSD card and file system as Figure 7-1



Heading address of user data area shall match with the heading of 64KB boundary of SD logical address.

Figure 7-1 Matching between logical address and file system

Note: Large Cluster unit is better for performance and WAF, for example, 128KB, 256KB or 512KB. Large cluster unit also can save write command numbers and few transfer time.

7.2. Basic Process of Error Handling

7.2.1. Retry Process

Execute the process by sending commands again, especially for signal issue between card and host.

7.2.2. Recovery Process

Confirm card status is in Transfer State, if card status is not in Transfer State, please issue Stop command to recover it and execute or continue flow. If there was UECC during read/write status, we could use recovery process to recover it.

7.2.3. Tuning Write Command Process

In order to adjust Host CMD and CLK timing, the way is issue tuning command to confirm what the device response and data was received by host. Based on the response, host was adjusting the timing step by step and recording the pass range. Through this flow host could adjust the appropriate timing settings to avoid unexpected handshaking issue.

7.2.4. Tuning Read Command Process

In order to adjust Host CLK and DAT timing, the way is issue tuning command to confirm what the device response and data was received by host. Based on the response, host was adjusting the timing step by step and recording the pass range. Through this flow host could adjust the appropriate timing settings to avoid unexpected handshaking issue.

7.2.5. Exception Handling Process

No doubt that sometimes we would face all error handling above could not recover it successfully, and we could react based on the situation.

- If there was error in response, we could re-initialize the card.
- If it was signal issue, we could set up signal status by reading data and tuning command.

7.3. Common Error Handling in SPI and SD mode

7.3.1. Time-out

Run the Retry Process. No response from CMD, it might be signal or status got problem. To avoid the infinite loop, implement a retry counter in the host so that, if the retry counter expires, the exception handling starts in the host.

7.3.2. Error Detect (CMD CRC Error)

Run the Recovery Process. If it got second time failure with CRC, the setting might be too margin to receive response stably. Suggestion is use tuning write command to fix timing and then retry it.

7.3.3. Error Detect (Other Error) in SPI and SD mode

Run the Recovery Process.

7.3.4. Others

Most errors could be recovered by running the Recovery Process, let card come into Transfer State and then

executing the flow we planned. If it does not work, please use exception method to come back initial state.

7.4. Data Error Handling in SPI and SD mode

7.4.1. Time-out

Run the Recovery Process. While the state was recovered, run the flow again.

7.4.2. Read CRC16 Error

Run the Recovery Process. If it got second time failure with CRC, the setting might be too margin to receive data stably. Suggestion is use tuning read date to fix timing and then retry it.

7.4.3. Write CRC Status Error

Run the Recovery Process. If it got second time failure with CRC, the setting might be too margin to receive CRC status stably. Suggestion is use tuning read date to fix timing and then retry it.

7.4.4. Others

Most errors could be recovered by running the Recovery Process, let card come into Transfer State and then executing the flow we planned.

7.5. Multiple Block Write (CMD25) Process

- If Response is ADDRESS_OUT_OF_RANGE, please confirm writing address.
- If Response is DEVICE_IS_LOCKED, please stop writing data.
- If Response is COM_CRC_ERROR, run retry or tuning.

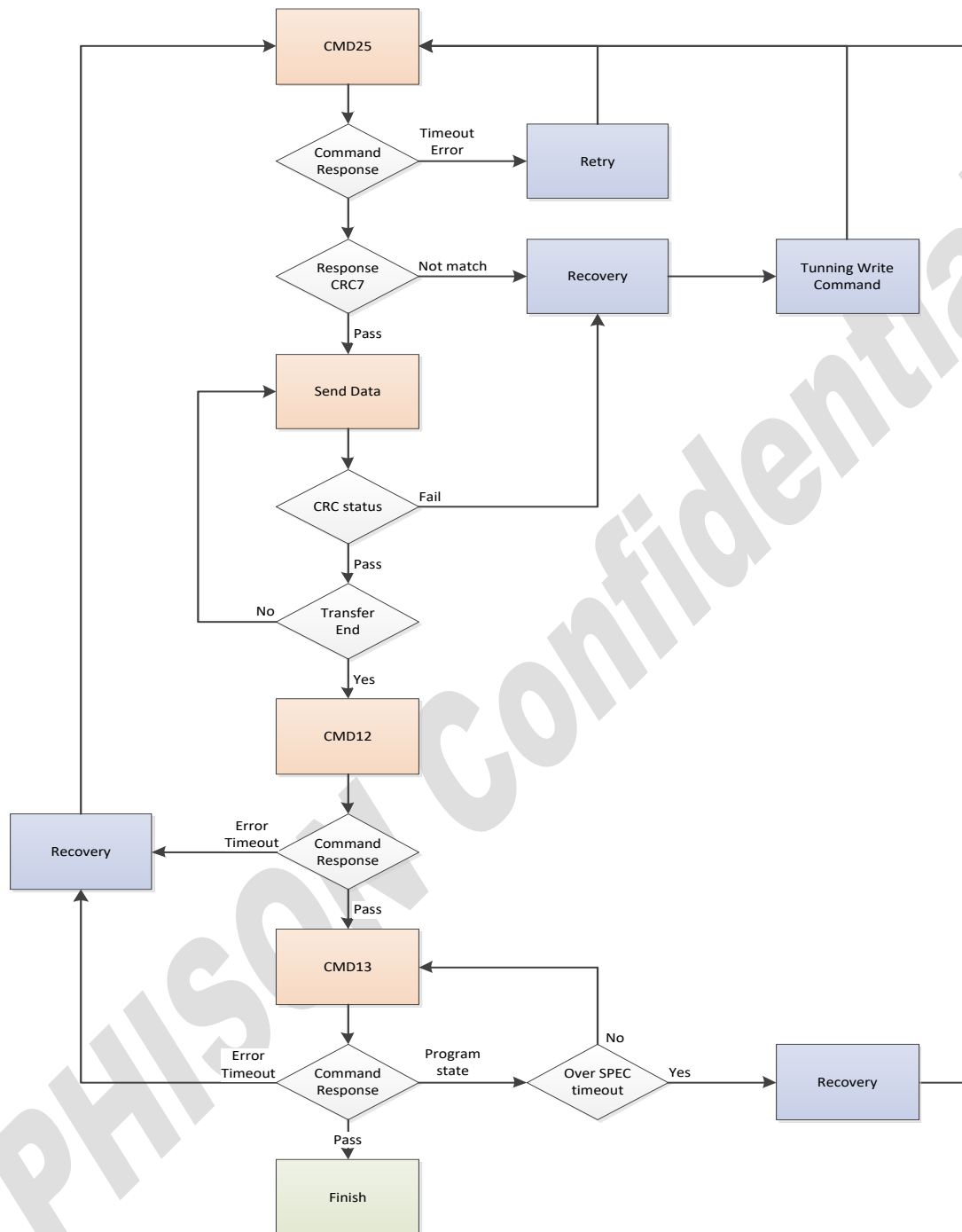


Figure 7-2 Multiple Write (CMD25) Error Handling

7.6. Retry Error handling

In order to avoid signal issue caused unexpected response from device, we could use Retry Process to fix it.

- Please make sure card state is in transfer state before issuing following commands.
- To avoid the infinite loop, implement a retry counter in the host.
- If the device could not respond to CMD13 normally, please run exception handling to recover card status.

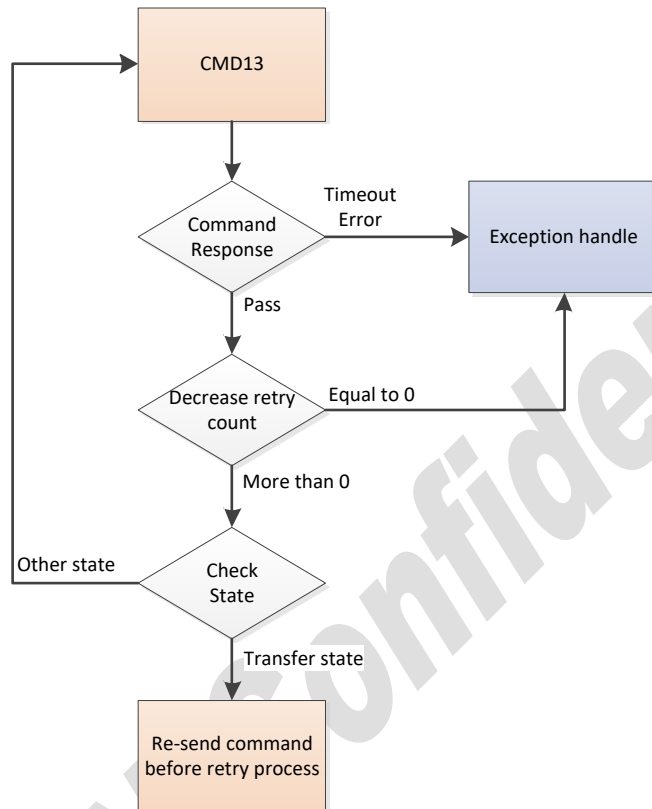


Figure 7-3 Retry Error Handling Process

7.7. Recovery Error Handling

Sometimes the device failure could not be recovered by Retry Process, it suggests to execute STOP Command (CMD12) to stop whole commands and response and then run following flow.

- Please confirm card status is in Transfer state
- In order to avoid infinite loops, host has to set up a retry counter number.

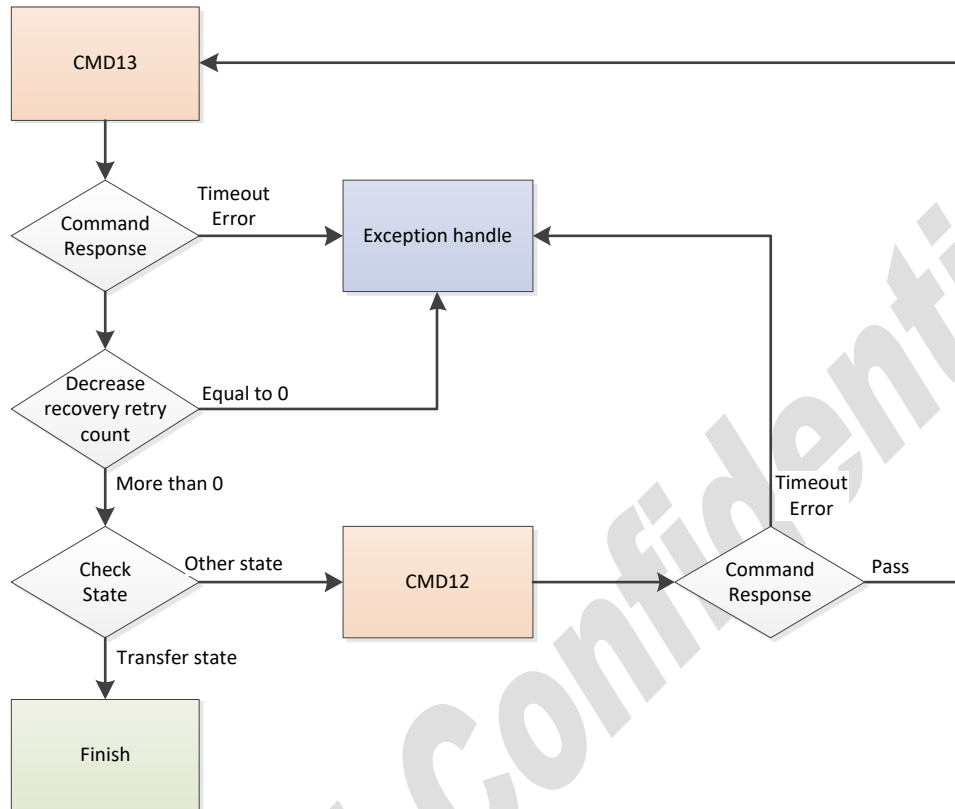


Figure 7-4 Recovery Error Handling Process

7.8. Tuning Write Command Error Handling

Reconfirm the card's pass range, to make sure card could receive host commands.

- If there was no any pass window, it might be connect issue or signal issue
- Pass Range depends on frequency level, higher frequency makes fewer pass range

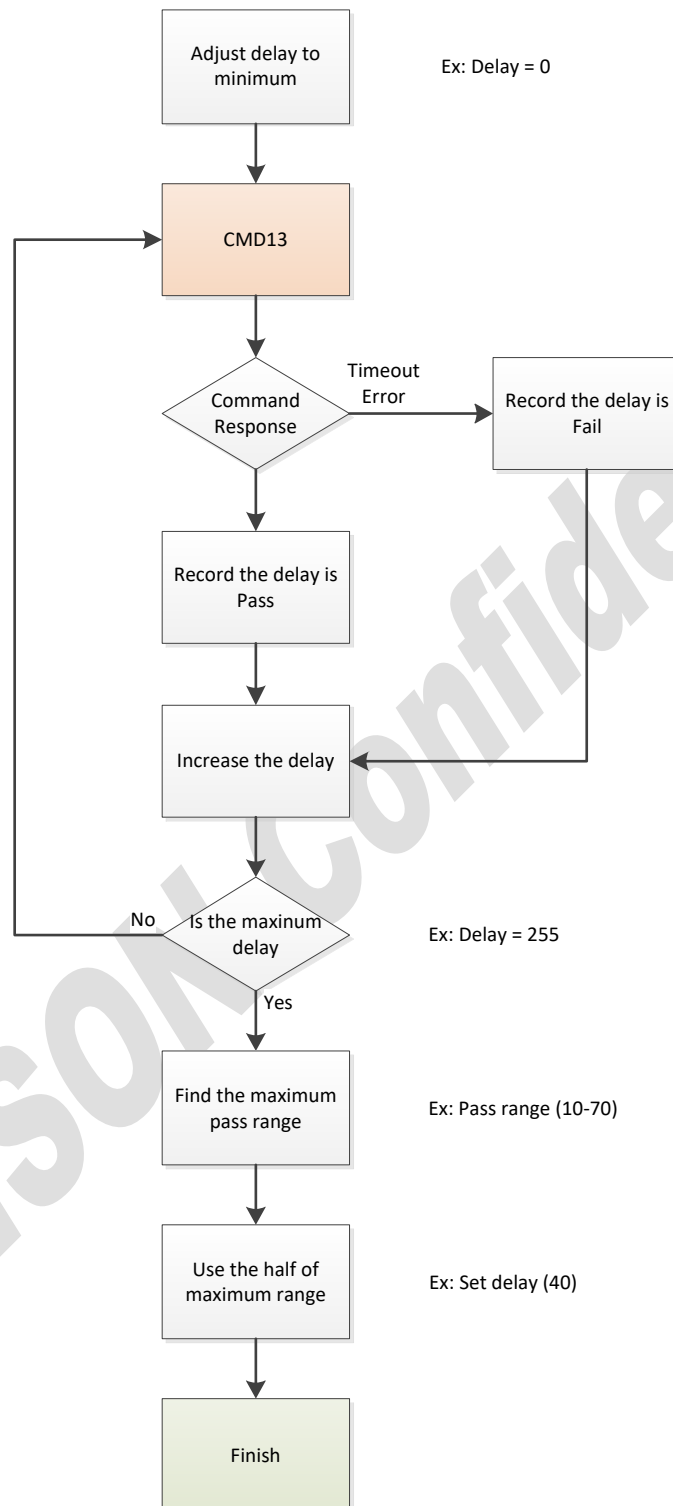


Figure 7-5 Tuning Write Command Error Handling Process

7.9. Exception Error Handling

- Error in Card's response or data output time-out, it could re-initialize the card.
- If there was CMD CRC7 issue, it could use tuning write command process to find out appropriate timing.
- If there was DAT CRC16 issue, it could use tuning read command process to find out appropriate timing.

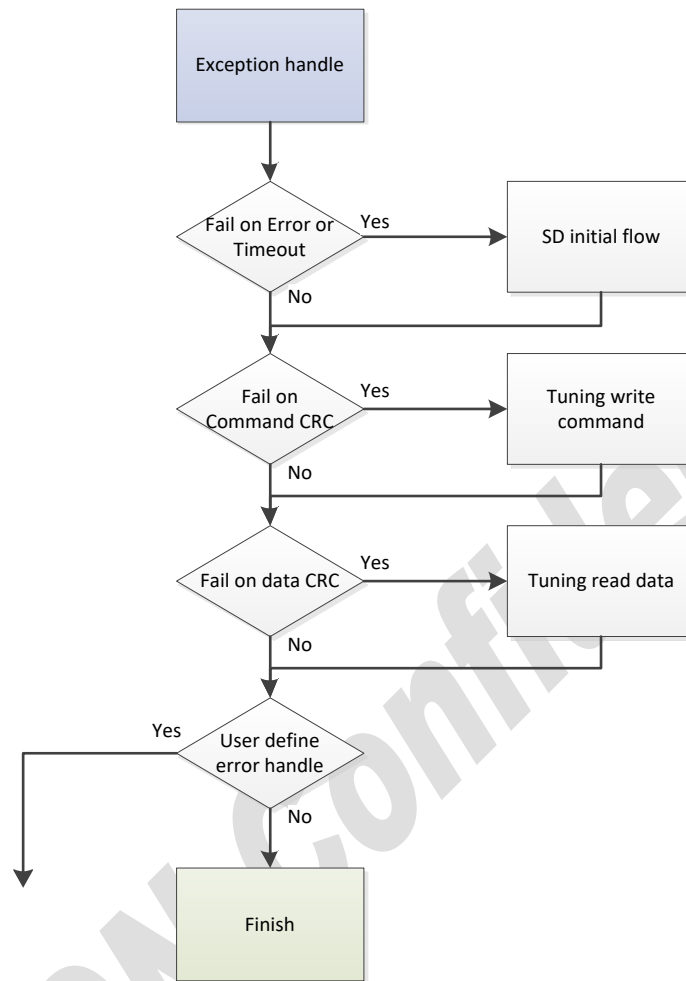


Figure 7-6 Exception Error Handling Process

7.10. Multiple Blocks Read (CMD18) Error Handling Process

- If card responded ADDRESS_OUT_OF_Range, please check writing address
- If card responded DEVICE_IS_LOCKED, please stop writing data
- If card responded COM_CRC_ERROR, run Retry or Tuning Process

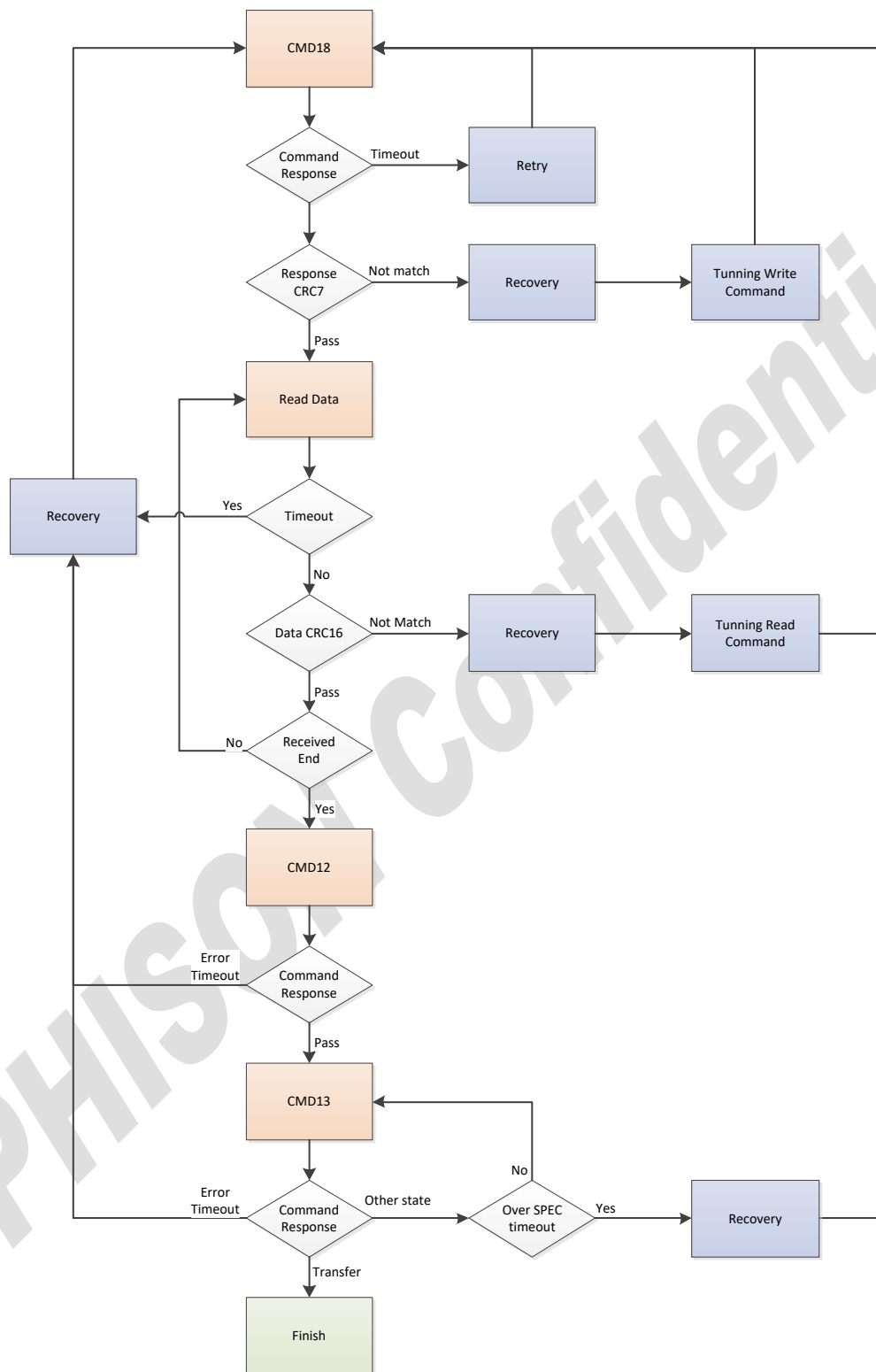


Figure 7-7 Multiple Blocks Read (CMD18) Error Handling Process

7.11. Tuning Read Data Error Handling

Reconfirm the card's pass range, to make sure host could receive card's Response and Data.

- If there was no any pass window, it might be connect issue or signal issue
- Pass Range depends on frequency level, higher frequency makes fewer pass range

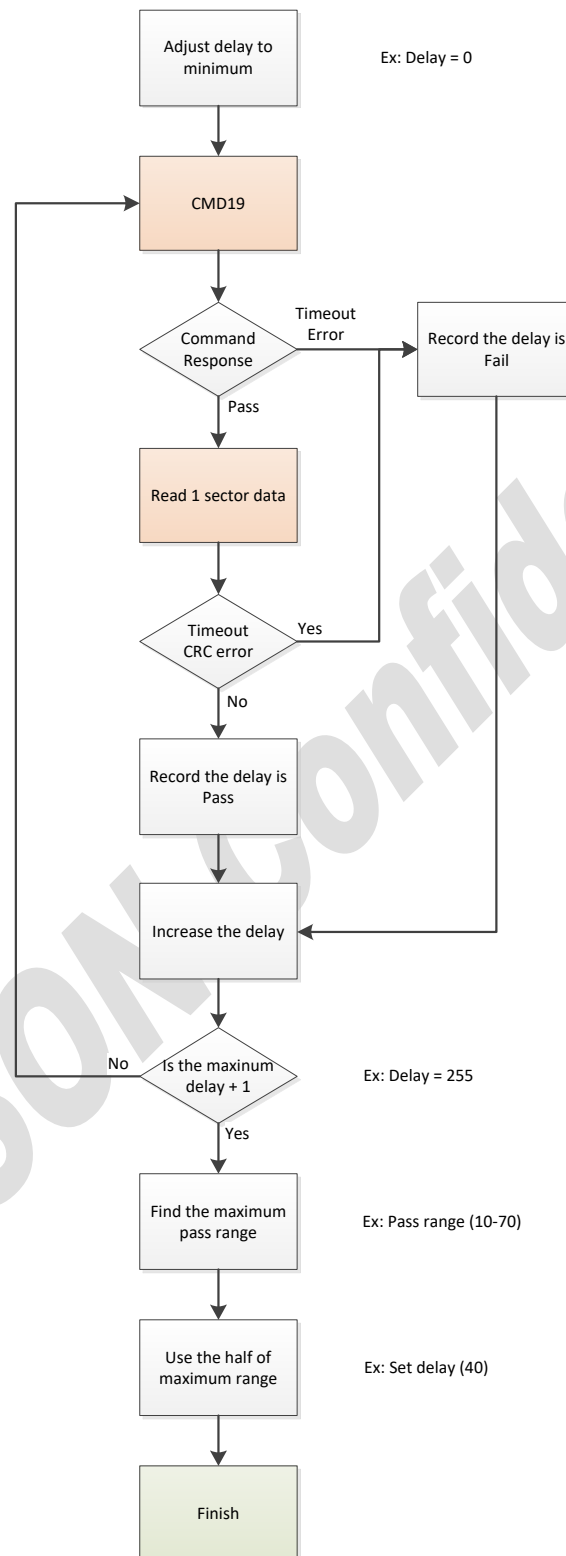


Figure 7-8 Tuning Read Data Error Handling Process

8. REGISTERS

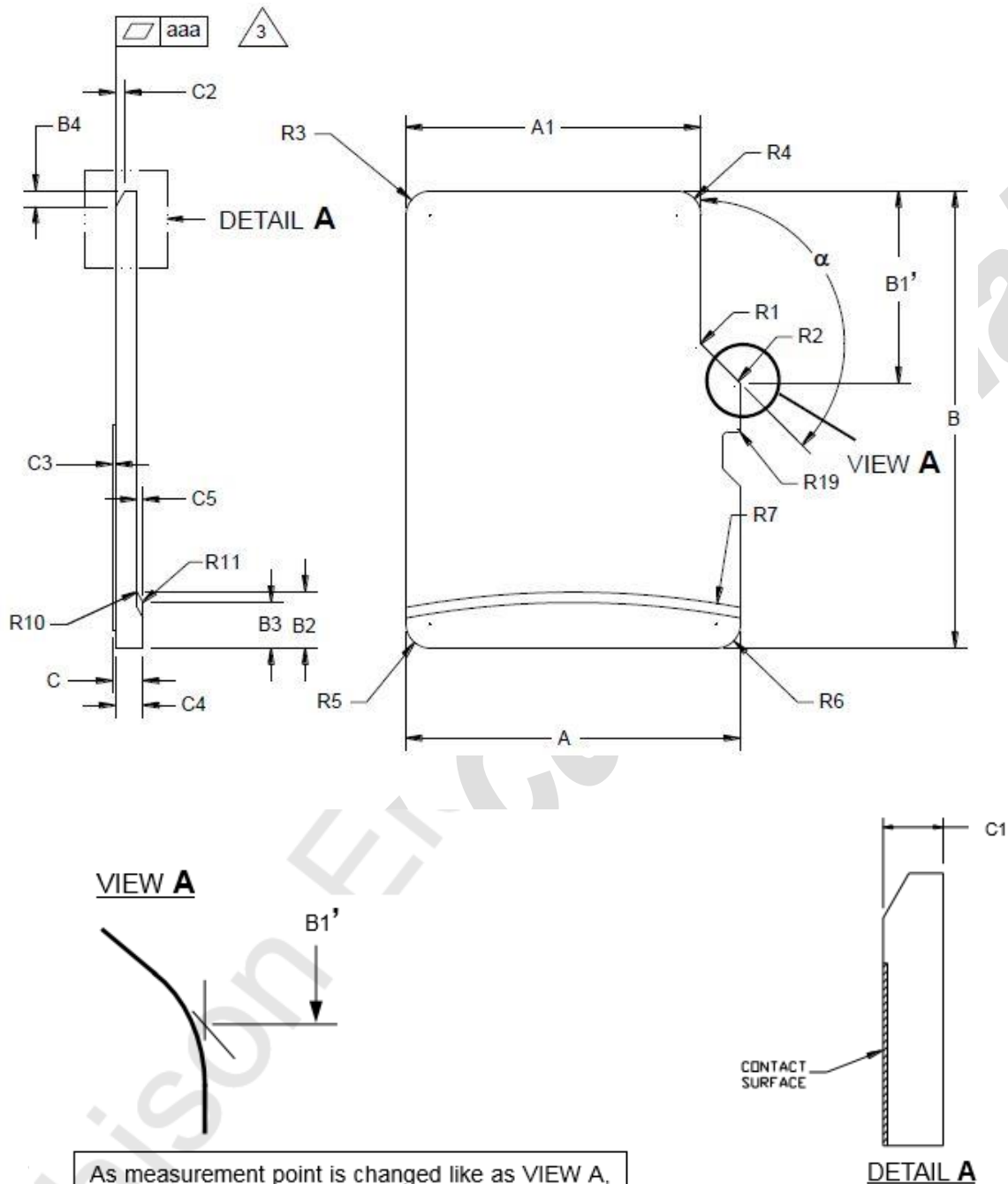


8.1. Card Registers

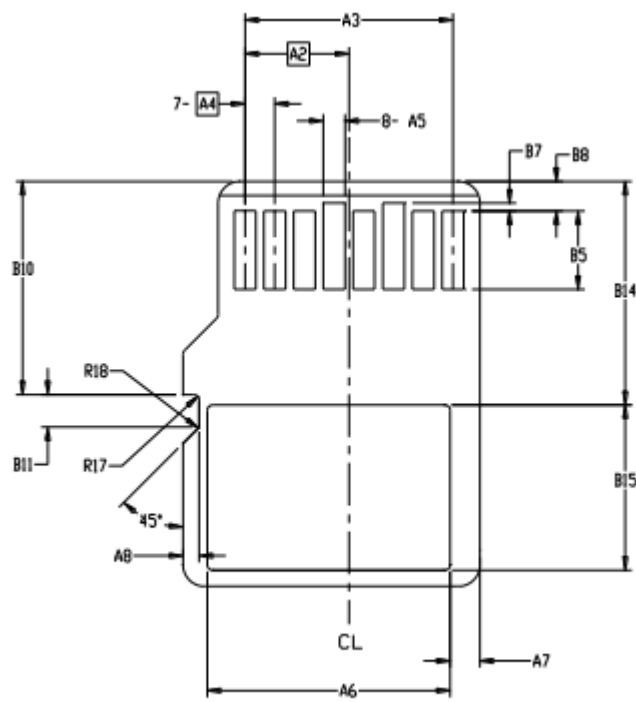
Name	Width	Description
CID	128bit	Card identification number; card individual number for identification. Mandatory
RCA ¹	16bit	Relative card address; local system address of a card, dynamically suggested by the card and approved by the host during initialization. Mandatory
DSR	16bit	Driver Stage Register; to configure the card's output drivers. Optional
CSD	128bit	Card Specific Data; information about the card operation conditions. Mandatory
SCR	64bit	SD Configuration Register; information about the SD Memory Card's Special Features capabilities Mandatory
OCR	32bit	Operation conditions register. Mandatory.
SSR	512bit	SD Status; information about the card proprietary features Mandatory
OCR	32bit	Card Status; information about the card status Mandatory

(1) RCA register is not used (or available) in SPI mode.

9. PHYSICAL DIMENSION



As measurement point is changed like as VIEW A, symbol B1 is changed to symbol B1'. There is no modification in mechanical dimension.



Type A

SYMBOL	COMMON DIMENSION			NOTE
	MIN	NOM	MAX	
A	10.90	11.00	11.10	
A1	9.60	9.70	9.80	
A2	-	3.85	-	BASIC
A3	7.60	7.70	7.80	
A4	-	1.10	-	BASIC
A5	0.75	0.80	0.85	
A6	-	-	8.50	
A7	0.90	-	-	
A8	0.60	0.70	0.80	
A9	0.80	-	-	
A10	1.35	1.40	1.45	
A11	6.50	6.60	6.70	
A12	0.50	0.55	0.60	
A13	0.40	0.45	0.50	
A14	0.05	-	-	
A15	5.71	5.81	5.91	
A16	6.47	6.57	6.67	
A17	6.62	6.72	6.82	
A18	7.38	7.48	7.58	
A19	7.75	7.85	7.95	
A20	8.55	8.65	8.75	
A21	0.90	-	-	
A22	-	-	8.50	
B	14.90	15.00	15.10	
B1'	6.13	6.23	6.33	
B2	1.64	1.84	2.04	
B3	1.30	1.50	1.70	
B4	0.42	0.52	0.62	
B5	2.80	2.90	3.00	
B6	5.50	-	-	
B7	0.20	0.30	0.40	
B8	1.00	1.10	1.20	
B9	-	-	9.00	
B10	7.80	7.90	8.00	
B11	1.10	1.20	1.30	
B12	3.60	3.70	3.80	
B13	2.80	2.90	3.00	
B14	8.20	-	-	
B15	-	-	6.20	
B16	5.80	5.90	6.00	
B17	0.20	0.30	0.40	
B18	7.80	8.80	8.90	
B19	8.70	8.80	8.90	
B20	-	3.20	-	REF
B21	1.90	2.00	2.10	
B22	9.00	-	-	
B23	0.10	-	-	

C	0.90	1.00	1.10	
C1	0.60	0.70	0.80	
C2	0.20	0.30	0.40	
C3	0.00	-	0.15	
C4	0.80	-	1.10	
C5	0.15	-	-	
D1	1.00	-	-	
D2	1.00	-	-	
D3	1.00	-	-	
R1	0.20	0.40	0.60	
R2	0.20	0.40	0.60	
R3	0.70	0.80	0.90	
R4	0.70	0.80	0.90	
R5	0.60	0.80	0.90	
R6	0.60	0.80	0.90	
R7	29.50	30.00	30.50	
R10	-	0.20	-	
R11	-	0.20	-	
R17	0.10	0.20	0.30	
R18	0.20	0.40	0.60	
R19	0.05	-	0.20	
R20	△	-	0.15	
α	133°	135°	137°	
aaa	-	-	0.10	

Notes:

1. DIMENSIONING and TOLERANCING per ASME Y14.5M-1994.
2. Dimensions are in millimeters.
3. COPLANARITY is additive to C1 MAX thickness.
4. All edges shall not be sharp as tested per UL1439 "Test for Sharpness of Edges on Equipment."
5. Refer to Appendix E about test method of warpage.
6. As measurement point is changed, symbol B1 is changed to symbol B1'.
7. C4 and C5 are added from Version 4.00.

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- (3) Phison's compliance with customer (including customer's suppliers, subcontractors or downstream customers) indicated instructions, technologies, designs, specifications, materials, components, parts;
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